Task 1 :

```
sigmoid@sigmoid-ThinkPad-T450:~/Projects$ sudo apt install postfix
[sudo] password for sigmoid:
Sorry, try again.
[sudo] password for sigmoid:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
postfix is already the newest version (3.8.6-1build2).
The following packages were automatically installed and are no longer required:
  kwayland-data kwayland-integration libdbusmenu-qt5-2 libdouble-conversion3
  libkf5config-bin libkf5config-data libkf5configcore5 libkf5coreaddons-data
  libkf5coreaddons5 libkf5notifications-data libkf5notifications5
  libkf5waylandclient5 libkf5windowsystem-data libkf5windowsystem5 libmd4c0
  libpcre2-16-0 libqt5core5t64 libqt5dbus5t64 libqt5gui5t64 libqt5network5t64
  libqt5qml5 libqt5qmlmodels5 libqt5quick5 libqt5svg5 libqt5texttospeech5
  libqt5waylandclient5 libqt5waylandcompositor5 libqt5widgets5t64
  libqt5x11extras5 libxcb-xinerama0 libxcb-xinput0 qt5-gtk-platformtheme
  qtspeech5-speechd-plugin qttranslations5-l10n qtwayland5
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 288 not upgraded.
```

```
sigmoid@sigmoid-ThinkPad-T450:~/Projects$ sudo systemctl status postfix
[sudo] password for sigmoid:
● postfix.service - Postfix Mail Transport Agent
     Loaded: loaded (/usr/lib/systemd/system/postfix.service; enabled; preset: enabled)
     Active: active (exited) since Sun 2026-03-01 21:49:25 IST; 3h 28min ago
       Docs: man:postfix(1)
   Main PID: 4021 (code=exited, status=0/SUCCESS)
        CPU: 5ms

Mar 01 21:49:25 sigmoid-ThinkPad-T450 systemd[1]: Starting postfix.service - Postfix Mail Transport Agent...
Mar 01 21:49:25 sigmoid-ThinkPad-T450 systemd[1]: Finished postfix.service - Postfix Mail Transport Agent.
sigmoid@sigmoid-ThinkPad-T450:~/Projects$
```

```
sigmoid@sigmoid-ThinkPad-T450:~/Projects$ echo "This is a test Mail" | mail -s "This is Test Subject" sigmoid
sigmoid@sigmoid-ThinkPad-T450:~/Projects$ mail
"/var/mail/sigmoid": 1 message 1 new
>N   1 Sigmoid            Sun Mar  1 21:52  13/500   This is
? 1
Return-Path: <sigmoid@sigmoid-ThinkPad-T450>
X-Original-To: sigmoid
Delivered-To: sigmoid@sigmoid-ThinkPad-T450
Received: by sigmoid-ThinkPad-T450 (Postfix, from userid 1000)
        id 14A3B9800BC; Sun,  1 Mar 2026 21:52:54 +0530 (IST)
Subject: This is Test Subject
To: sigmoid@sigmoid-ThinkPad-T450
User-Agent: mail (GNU Mailutils 3.17)
Date: Sun,  1 Mar 2026 21:52:54 +0530
Message-Id: <20260301162254.14A3B9800BC@sigmoid-ThinkPad-T450>
From: Sigmoid <sigmoid@sigmoid-ThinkPad-T450>

This is a test Mail
?
```

Task 2

```
sudo useradd -m -s /bin/bash Kunal
```

```
sigmoid@sigmoid-ThinkPad-T450:/$ sudo passwd Kunal
New password:
Retype new password:
passwd: password updated successfully
```

```
sigmoid@sigmoid-ThinkPad-T450:/$ groups Kunal
Kunal : Kunal
sigmoid@sigmoid-ThinkPad-T450:/$ su - Kunal
Password:
```

```
Kunal@sigmoid-ThinkPad-T450:~$ pwd
/home/Kunal
Kunal@sigmoid-ThinkPad-T450:~$ sudo apt update
[sudo] password for Kunal:
Kunal is not in the sudoers file.
This incident has been reported to the administrator.
Kunal@sigmoid-ThinkPad-T450:~$
```

Task 3 :

```
  GNU nano 7.2                /usr/local/bin/describe
#!/bin/bash
ls

                              [ Read 2 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify
```

```
root@sigmoid-ThinkPad-T450:/# sudo nano /usr/local/bin/describe
root@sigmoid-ThinkPad-T450:/# sudo chmod +x /usr/local/bin/describe
root@sigmoid-ThinkPad-T450:/# describe
bin                home                mnt   sbin.usr-is-merged  usr
bin.usr-is-merged  lib                 opt   snap                var
boot               lib64               proc  srv
cdrom              lib.usr-is-merged   root  swap.img
dev                lost+found          run   sys
etc                media               sbin  tmp
root@sigmoid-ThinkPad-T450:/#
```

```
root@sigmoid-ThinkPad-T450:/# cd /home/
root@sigmoid-ThinkPad-T450:/home# describe
aryan  babubhai  Kunal  salman  sigmoid
root@sigmoid-ThinkPad-T450:/home# cd /etc/
```

```
root@sigmoid-ThinkPad-T450:/# cd /home/
root@sigmoid-ThinkPad-T450:/home# describe
aryan  babubhai  Kunal  salman  sigmoid
root@sigmoid-ThinkPad-T450:/home# cd /etc/
root@sigmoid-ThinkPad-T450:/etc# describe
adduser.conf            hdparm.conf          printcap
aliases                 host.conf            profile
aliases.db              hostname             profile.d
alsa                    hosts                protocols
alternatives            hosts.allow          pulse
anacrontab              hosts.deny           python3
apache2                 hp                   python3.12
apg.conf                ifplugd              rc0.d
apm                     init                 rc1.d
apparmor                init.d               rc2.d
apparmor.d              initramfs-tools      rc3.d
apport                  inputrc              rc4.d
apt                     insserv.conf.d       rc5.d
at.deny                 ipp-usb              rc6.d
```

```
root@sigmoid-ThinkPad-T450:/etc# su - Kunal
Kunal@sigmoid-ThinkPad-T450:~$ describe
Kunal@sigmoid-ThinkPad-T450:~$ ls
Kunal@sigmoid-ThinkPad-T450:~$ touch file1.txt
Kunal@sigmoid-ThinkPad-T450:~$ mkdir Documents
Kunal@sigmoid-ThinkPad-T450:~$ mkdir Notes
Kunal@sigmoid-ThinkPad-T450:~$ touch user.txt
Kunal@sigmoid-ThinkPad-T450:~$ describe
Documents  file1.txt  Notes  user.txt
Kunal@sigmoid-ThinkPad-T450:~$
```

Task 4 :

```
echo “This is research data” > research.txt
```

gzip research.txt

Then move that file into another path

sudo mv research.txt.gz /var/log/research.txt.gz

```
root@sigmoid-ThinkPad-T450:/# find / -name "research*" 2>/dev/null
/var/log/research.txt.gz
root@sigmoid-ThinkPad-T450:/# file /var/log/research.txt.gz
/var/log/research.txt.gz: gzip compressed data, was "research.txt", last modified: Tue Mar  3 14:09:38 2026,
from Unix, original size modulo 2^32 22
root@sigmoid-ThinkPad-T450:/# gunzip /var/log/research.txt.gz
root@sigmoid-ThinkPad-T450:/# ls
```

```
root@sigmoid-ThinkPad-T450:/# find / -name "research*" 2>/dev/null
/var/log/research.txt
root@sigmoid-ThinkPad-T450:/# cat /var/lo
local/ lock/  log/
root@sigmoid-ThinkPad-T450:/# cat /var/log/research.txt
This is research data
root@sigmoid-ThinkPad-T450:/#
```

Task 5 :

sudo nano /etc/profile

```
  GNU nano 7.2                    /etc/profile
# /etc/profile: system-wide .profile file for the Bourne shell (sh(1))
# and Bourne compatible shells (bash(1), ksh(1), ash(1), ...).

if [ "${PS1-}" ]; then
  if [ "${BASH-}" ] && [ "$BASH" != "/bin/sh" ]; then
    # The file bash.bashrc already sets the default PS1.
    # PS1='\h:\w\$ '
    if [ -f /etc/bash.bashrc ]; then
      . /etc/bash.bashrc
    fi
  else
    if [ "$(id -u)" -eq 0 ]; then
      PS1='# '
    else
      PS1='$ '
    fi
  fi
fi

if [ -d /etc/profile.d ]; then
  for i in /etc/profile.d/*.sh; do
    if [ -r $i ]; then
      . $i
    fi
  done
  unset i
fi
umask 777
```

```
sigmoid@sigmoid-ThinkPad-T450:~/LinuxAssignment$ umask
0002
sigmoid@sigmoid-ThinkPad-T450:~/LinuxAssignment$ sudo nano /etc/profile
sigmoid@sigmoid-ThinkPad-T450:~/LinuxAssignment$ umask 777
sigmoid@sigmoid-ThinkPad-T450:~/LinuxAssignment$ touch testfile.txt
sigmoid@sigmoid-ThinkPad-T450:~/LinuxAssignment$ ls -l testfile.txt
---------- 1 sigmoid sigmoid 0 Mar  3 23:58 testfile.txt
```

```
root@sigmoid-ThinkPad-T450:/# su - Kunal
Kunal@sigmoid-ThinkPad-T450:~$ ls
Documents  file1.txt  Notes  user.txt
Kunal@sigmoid-ThinkPad-T450:~$ echo "This is demofile" > demofile.txt
Kunal@sigmoid-ThinkPad-T450:~$ ls -l demofile.txt
---------- 1 Kunal Kunal 17 Mar  4 00:08 demofile.txt
Kunal@sigmoid-ThinkPad-T450:~$
```

Task 6 :

```
  GNU nano 7.2                    /usr/local/bin/showtime.sh
#!/bin/bash
while true; do
        echo "Current Time : $(date)" >> /root/showtime.log
        sleep 60
done
```

```
  GNU nano 7.2          /etc/systemd/system/showtime.service *
[Unit]
Description=Showtime Service - Prints current time every minute

[Service]
ExecStart=/usr/local/bin/showtime.sh
Restart=always
User=root

[Install]
WantedBy=multi-user.target
```

```
root@sigmoid-ThinkPad-T450:~# systemctl daemon-reload
root@sigmoid-ThinkPad-T450:~# systemctl enable showtime
root@sigmoid-ThinkPad-T450:~# service showtime start
root@sigmoid-ThinkPad-T450:~# service showtime status
● showtime.service - Showtime Service - Prints current time every minute
     Loaded: loaded (/etc/systemd/system/showtime.service; enabled; preset>
     Active: active (running) since Wed 2026-03-04 02:05:56 IST; 2min 27s >
   Main PID: 45939 (showtime.sh)
      Tasks: 2 (limit: 9193)
     Memory: 768.0K (peak: 1.4M)
        CPU: 17ms
     CGroup: /system.slice/showtime.service
             ├─45939 /bin/bash /usr/local/bin/showtime.sh
             └─46327 sleep 60

Mar 04 02:05:56 sigmoid-ThinkPad-T450 systemd[1]: Started showtime.service>
lines 1-12/12 (END)
```

```
Mar 04 02:05:56 sigmoid-ThinkPad-T450 systemd[1]: Started showtime.service>
root@sigmoid-ThinkPad-T450:~# cat /root/showtime.log
Current Time : Wed Mar  4 01:33:43 AM IST 2026
Current Time : Wed Mar  4 01:37:46 AM IST 2026
Current Time : Wed Mar  4 01:40:00 AM IST 2026
Current Time : Wed Mar  4 02:05:56 AM IST 2026
Current Time : Wed Mar  4 02:06:56 AM IST 2026
Current Time : Wed Mar  4 02:07:56 AM IST 2026
Current Time : Wed Mar  4 02:08:56 AM IST 2026
root@sigmoid-ThinkPad-T450:~#
```

```
root@sigmoid-ThinkPad-T450:~# service showtime stop
root@sigmoid-ThinkPad-T450:~# service showtime status
○ showtime.service - Showtime Service - Prints current time every minute
     Loaded: loaded (/etc/systemd/system/showtime.service; enabled; preset>
     Active: inactive (dead) since Wed 2026-03-04 02:10:07 IST; 6s ago
   Duration: 4min 10.476s
    Process: 45939 ExecStart=/usr/local/bin/showtime.sh (code=killed, sign>
   Main PID: 45939 (code=killed, signal=TERM)
        CPU: 26ms

Mar 04 02:05:56 sigmoid-ThinkPad-T450 systemd[1]: Started showtime.service>
Mar 04 02:10:07 sigmoid-ThinkPad-T450 systemd[1]: Stopping showtime.servic>
Mar 04 02:10:07 sigmoid-ThinkPad-T450 systemd[1]: showtime.service: Deacti>
Mar 04 02:10:07 sigmoid-ThinkPad-T450 systemd[1]: Stopped showtime.service>
lines 1-12/12 (END)
```